

A Bayesian approach for understanding the role of ship speed in whale–ship encounters

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Abstract. Mandatory or voluntary reductions in ship speed are a common management strategy for reducing deleterious encounters between large ships and large whales. This has produced strong resistance from shipping and marine transportation entities, in part because very few studies have empirically demonstrated whether or to what degree ship speed influences ship–whale encounters. Here we present the results of four years of humpback whale sightings made by observers aboard cruise ships in Alaska, representing 380 cruises and 891 ship–whale encounters. Encounters occurred at distances from 21 m to 1000 m ($\bar{x}$ = 567 m) with 61 encounters (7%) occurring between 200 m and 100 m, and 19 encounters (2%) within 100 m. Encounters were spatially aggregated and highly variable across all ship speeds. Nevertheless a Bayesian change-point model found that the relationship between whale distance and ship speed changed at 11.8 knots (6.1 m/s) with whales encountering ships, on average, 114 m closer when ship speeds were above 11.8 knots. Binning encounter distances by 1-knot speed increments revealed a clear decrease in encounter distance with increasing ship speed over the range of 7–17 knots (3.6–8.7 m/s). Our results are the first to demonstrate that speed influences the encounter distance between large ships and large whales. Assuming that the closer ships come to whales the more likely they are to be struck, our results suggest that reduced ship speed may be an effective management action in reducing the probability of a collision.

Key words: Alaska; Bayesian; change-point model; cruise ship; Glacier Bay National Park and Preserve, Alaska, USA; humpback whale; ship speed; ship strike; ship–whale encounters.

INTRODUCTION

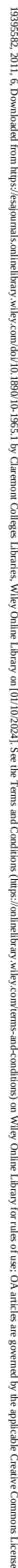
Understanding the interactions between large ships and large whales has become a global conservation issue owing to the dramatic increase in commercial shipping traffic in recent decades (Andrew et al. 2002, Ross 2005) and an emerging awareness of the deleterious impacts that large vessels may have on individuals and populations of large whales. For example, shipping produces underwater sound that may disrupt vital activities for whales, such as feeding or reproduction, or hinder communication (NRC 2005, Parks et al. 2011). Whale–ship encounters may also result in lethal or sublethal collisions (Laist et al. 2001, Kraus et al. 2005). Consequently, a number of U.S. and international management entities have initiated efforts aimed at reducing these impacts, focusing primarily on reducing spatial overlap between ships and whales (e.g.,

Fonnesbeck et al. 2008, Vanderlaan et al. 2008). In some cases, this management approach is both feasible and effective; in the Bay of Fundy, for example, slight changes in routing of vessels can reduce the relative risk of collision by up to 62% (Vandelaar et al. 2008). In other cases, however, this management action may not be effective or appropriate, either because whale distribution may shift over time, or because ships may not be able to reroute around high-use whale habitat when approaching ports of call, or in narrow passages. In these instances, the primary management action has been to implement regulations for, or request voluntary compliance with, reductions in ship speed (National Park Service 2006, NMFS 2008, Carrillo and Ritter 2009).

Although relatively common, there is considerable uncertainty in the effectiveness of regulating ship speed in meeting whale-conservation objectives due, in part, to a paucity of data (e.g., Vanderlaan and Taggart 2007, NMFS 2008). The difficulty of applying an experimental framework, and the legal and logistical issues associated with putting observers aboard large, ocean-ranging

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In some cases, however, the whale dove before the rangefinder binoculars could be used. For these encounters (31% of total) the distances were estimated. To determine if estimating the distance biased our results, on 10 different occasions during each cruise, the observers estimated the distance to inanimate objects in the water (e.g., logs, ice bergs) and then immediately used the rangefinder binoculars to record the actual distances. This “estimation error” (difference between actual and estimated) was relatively small (average is $\pm 12\%$ of the actual encounter distance across all distances) and unbiased (percentage error did not change appreciably across encounter distances). Thus, no corrections were made for estimated vs. observed distances.

Following each cruise, all encounter data were entered into a Microsoft Access database, and all spatial data were downloaded from the GPS units (Garmin, Olathe, Kansas, USA) using DNR Garmin (free software from the Minnesota Department of Natural Resources, *available online*).⁸ The cruise-ship track data (locations automatically logged every 5 s) and whale-sighting data (waypoints taken during whale observations) were merged in ArcGIS using a Visual Basic script. Ship speed at the time of whale sighting was calculated by extracting 10 points from the track data—five ship location points immediately preceding and immediately following a whale-encounter waypoint—and summing the distance over which the ship traveled during that time and dividing by the time it took to cover that distance.

Our analytic approach was to calculate the management-relevant metric of distance of the whale to the bulbous bow (where most whale-ship collisions are likely to occur) based on the distance of the observer to the whale and the bearing of the whale to the ship. This equates to calculating the length of the line segment AC based on the triangle $\triangle ABC$ with vertices of whale location (A), location of observer (B), and the bulbous bow (C). To do so, we first calculated the distance of the observer to the bulbous bow (BC), which is $\sqrt{h^2 + q^2}$ where h is the height of the observer above water and q is the distance of the bow (observer) to the bulbous bow at the water level (both of these ship dimensions were provided by the cruise companies). We then calculated the distance of the whale to the observer at water level $p = \sqrt{AB^2 - h^2}$. The encounter distance (AC) = $\sqrt{p^2 + q^2 - 2pq \cos \alpha}$ where α is the bearing of the whale relative to ship direction (180° when the whale is directly in front of the ship).

If several locations were taken on the same whale, only one encounter (the minimum distance) was used for analysis to ensure independence and because minimum encounter distance was the management-relevant metric. The maximum encounter distance was limited to <1 km for three reasons. First, it was unclear at what maximum distance whale-ship interactions can still be considered biologically meaningful. Second, we felt confident that using only encounters within 1 km would ensure a near 100% detection probability, similar to what was found by Zerbini et al. (2009) using distance sampling during ship-based abundance estimates for humpback whales in the Gulf of Alaska. Nevertheless, to test this assumption on nine separate cruises we placed two independent observers (double sampling) aboard the same cruise ship, one each on the port and starboard sides of the bow, and found 100% concurrence of sighting frequency. Finally, ships are almost always at least 2 km from shore in all directions while in our study area, although in a few areas they travel to near 1 km of shore (Fig. 1). Thus, by truncating encounters to 1 km, we assumed an

unbiased probability of detecting whales around the ship independent of geographic location.

We also divided the study area a priori into five separate subareas (Fig. 1) corresponding to differences in oceanographic conditions (Etherington et al. 2007). These subareas also reflected differences in whale habitat use, based on long-term monitoring efforts conducted by NPS personnel (Neilson and Gabriele 2005). No oceanographic monitoring has occurred in the West Icy subarea.

To test whether there was a change in the mean encounter distance as a function of ship speed (given the management application, the knot is the speed metric used in this analysis), we used a change-point model. Change-point models have the flexibility to identify changes in a relationship while incorporating uncertainty (Thomson et al. 2010). Our change-point model can be described as

$$Y_i \sim \log \mathcal{N}(\beta_k, \sigma^2) \quad (1)$$

$$k = 1 + \text{step}(X_i - X_{CP})$$

where the Y_i are the individual encounters (each with a distance and speed) $i = 1, \dots, N$, which are modeled by the mean values β_k and variance σ^2 , and X_i is the speed during encounters $i = 1, \dots, N$. Assuming a single change point (CP) is identified, the mean encounter distance can have two levels above and below X_{CP} , thus $k = 1, 2$. The mean level is determined by whether the speed of the vessel is above or below the threshold with mean distance β_1 and β_2 , respectively. The function $\text{step}() = 1$ if the value inside the parenthesis is positive and equates to 0 if negative. There are four coefficients to be estimated in the change-point model: β_1 , β_2 , σ^2 , and X_{CP} . Of particular interest is how the speed threshold X_{CP} affects parameter estimates of β_1 and β_2 . We used a log-normal distribution in the change-point analysis because all encounter distances were positive.

The change-point model was conducted in a Bayesian estimation framework that allowed us to calculate probability densities representing the uncertainty for each of the four model parameters. We used non-informative prior probabilities for all model coefficients with the goal of allowing the data to support identification of the speed threshold (change point) at which there was a difference in mean distance from ship to whale. This noninformative approach corresponded to the change-point coefficient having a uniform prior distribution between 9 and 17 knots (4.6 and 8.7 m/s) [$X_{CP} \sim \text{Unif}(9, 17)$]. Although encounters with whales occurred at speeds faster and slower than this range, these speeds corresponded to the range where we had at least 20 observations in each 1-knot speed increment (1 knot = 0.5144 m/s). The mean distances above and below the change point were specified with a lognormal distribution with large variance [$\beta_1 \sim \log \mathcal{N}(0, 1000)$, $\beta_2 \sim \log \mathcal{N}(0, 1000)$] and the standard deviation of the lognor-

⁸<http://www.dnr.state.mn.us/mis/gis/tools/arcview/extensions/DNRCGarmin/DNRCGarmin.html>

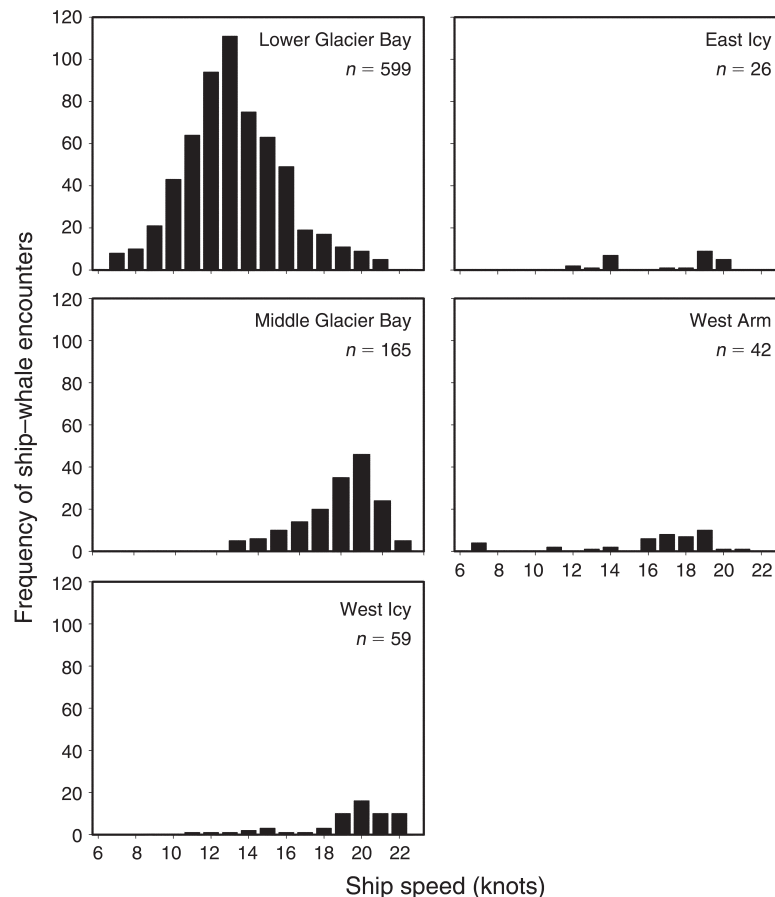


FIG. 2. Frequency of encounters between cruise ships and humpback whales, 2006–2009, by subarea and binned by 1-knot (0.5144 m/s) ship-speed increments ($n = 891$ unique ship-whale encounters).

mal observations was specified with a diffuse uniform distribution [$\sigma \sim \text{Unif}(0, 20)$]. The coefficients were estimated through use of the Gibbs sampler, a Markov chain Monte Carlo (MCMC) algorithm implemented in WinBUGS (Spiegelhalter et al. 2003). Multiple chains were run using dispersed initial values for each model to ensure the MCMC chain converged to a stationary target distribution. Monitored parameters in all models had scale reduction factor (SRF) values that indicated samples were being drawn from the target distribution (i.e., $\text{SRF} \sim 1$). The initial 30% of the samples were used to reach the stationary target distribution and were discarded with the subsequent samples thinned to produce approximately 1000 draws from the stationary target distributions. The 1000 draws were used to compute the posterior mean and 95% central probability intervals (credible intervals; 95% CrI). The diagnostics were implemented using the R2WinBUGS package in R (R Development Core Team 2007).

RESULTS

From 2006 through 2009 observers logged more than 2760 hours recording ship-whale encounters during 380 unique ship entries into Glacier Bay, Alaska, USA.

Observations occurred aboard 18 different cruise ships and constituted ~49% of all ship entries into the park during that period. A total of 891 unique ship-whale encounters were recorded at distances ranging from 21 m to 1000 m ($\bar{x}$ = 567 m, median = 576 m). Most (811) of the encounters (91%) occurred between 300 m and 1000 m although there were 61 encounters (7%) at 200–100 m, and 19 encounters (2%) within 100 m. No collisions were detected. Nearly all (754) of the encounters were with a single whale (85%) or with a group of two whales (n = 96 encounters; 11%); the remaining 4% of encounters occurred with group sizes of ≥ 3 whales. Encounter distances did not differ when comparing group sizes of 1 vs. >1 (group size 1 whale, $\bar{x}$ = 565 ± 10 m; group size ≥ 2 whales, $\bar{x}$ = 572 ± 29 m; t = 1.96, P = 0.81) and thus all analyses were considered independent of group size.

The average and range of ship speed (over ground) during encounters, as well as the frequency of encounters varied dramatically among subareas (Fig. 2). For example, in Middle Glacier Bay there was a large number of encounters recorded (165 encounters) but mostly (>76%) between 17 and 20 knots (8.7 and 10.3 m/s). In contrast, there was high variation in the speed when whales were encountered in the West Arm subarea

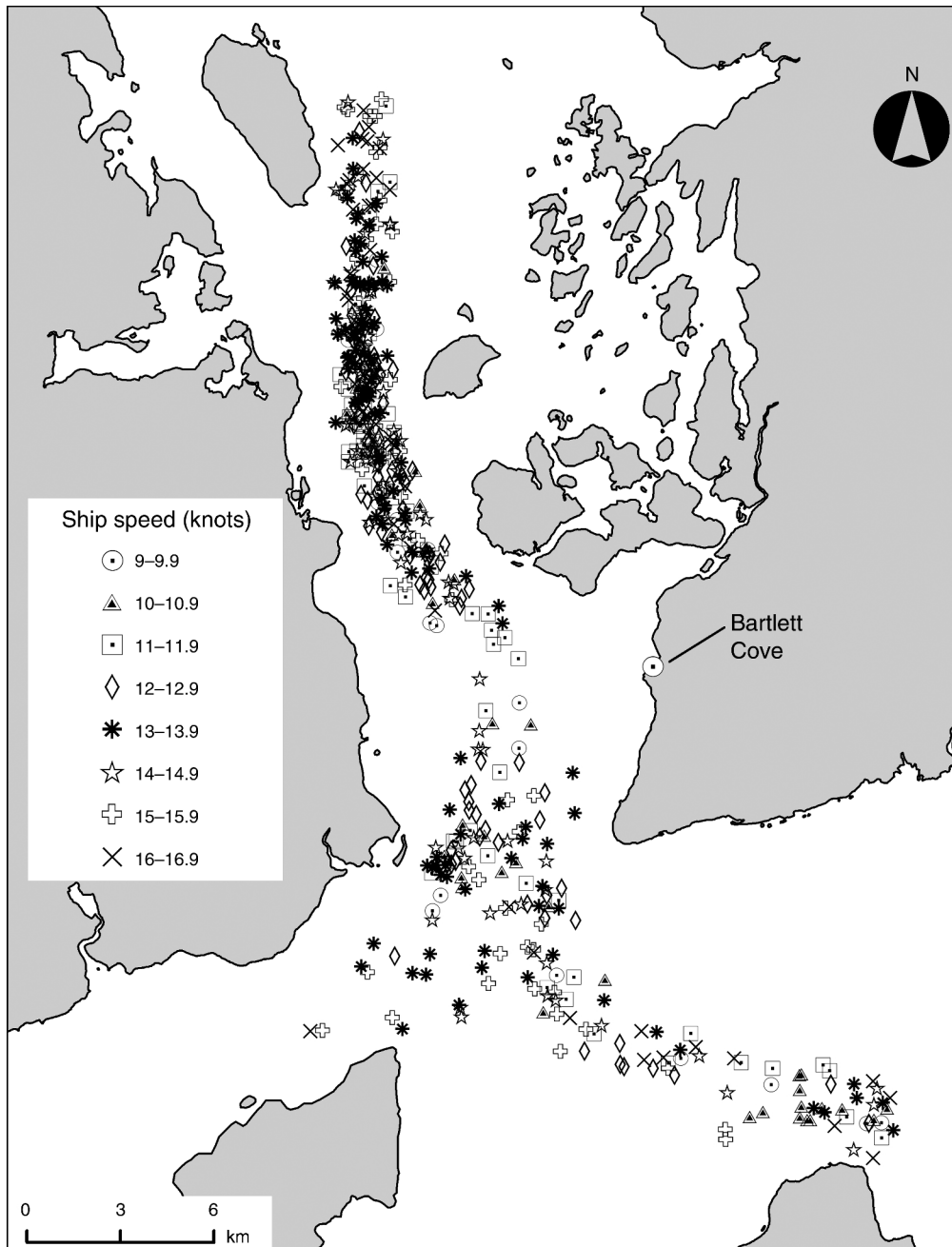


FIG. 3. Locations of humpback whales ($n = 599$) during encounters with cruise ships at different ship speeds in the Lower Glacier Bay subarea, 2006–2009. For ship speeds, note that 1 knot = 0.5144 m/s.

but very low sample size (<10 encounters) for any given speed. In contrast there were at least 20 encounters at speeds from 9 to 17 knots (4.6 and 8.7 m/s) in the Lower Glacier Bay subarea. This was due to ships slowing during transfer of NPS interpretive rangers from Bartlett Cove, speed regulations (with variable compliance), and a consistent presence of whales.

Thus, to reduce the possibility of spatial autocorrelation between speed and probability of encounter, we used only the data from the Lower Glacier Bay subarea

in our Bayesian change-point model. Doing so reduced our sample size only marginally because this subarea accounted for almost 70% of all encounters recorded during the study. Within the Lower Glacier Bay sub-area, there was no evidence that whale encounters were spatially aggregated at higher ship speeds (Fig. 3).

The Bayesian change-point model identified a high probability of a change in the relationship between ship speed and encounter distance around 11.8 knots (6.1 m/s) (Fig. 4A). The maximum probability of a change

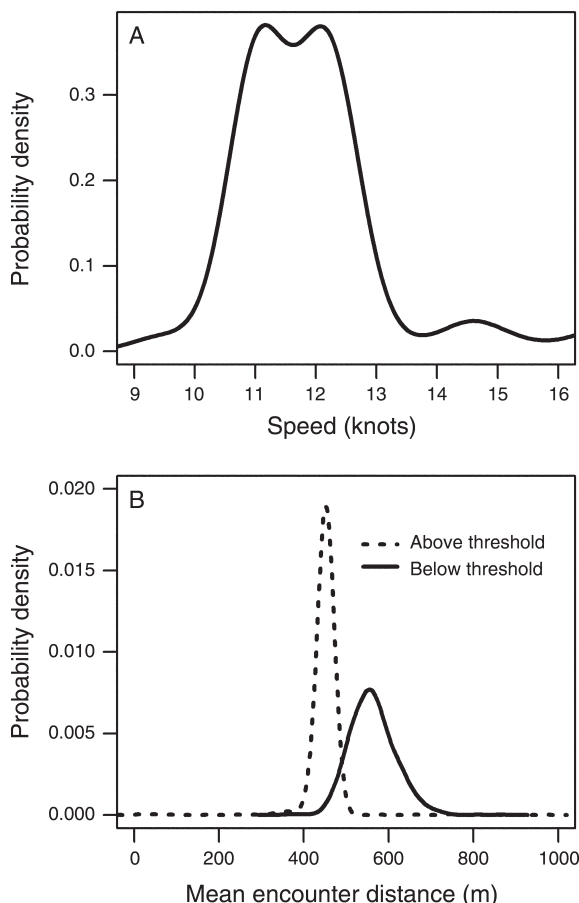


FIG. 4. Posterior probability distribution of (A) the speed threshold coefficient in the Bayesian change-point model (1 knot = 0.5144 m/s) and (B) the mean encounter distances to whales above (solid line) and below (dashed line) the speed threshold.

point occurred at 11.8 knots although there was only a slightly lower probability the change point occurred at 12.2 knots (6.3 m/s) (Fig. 4A). The probability that the change point occurred over the interval between 10 to 13 knots (5.1–6.9 m/s) was 0.88, i.e., $P(10 \text{ knots} < X_{CP} < 13 \text{ knots}) = 0.88$. As a result, the highest probability for β_1 and β_2 , the average encounter distances above and below the 11.8 knot change point were 448 m (95% credible interval, 398, 485) and 562 m (95%CrI, 468, 676; Fig. 4B). Binning the speeds by 1-knot (0.5144 m/s) increments clearly demonstrates that, on average, when ships were traveling faster, whales were encountered more closely (Fig. 5).

DISCUSSION

Reducing the probability of severe injury or death as a result of a collision—a primary management objective in regulating ship speed—can occur by reducing the probability of any of the series of constituent events during a collision including the probability of (1) an encounter given a ship's presence in an area, (2) a close

(severe) encounter given an encounter, (3) a collision given a severe encounter, and (4) severe injury or death given a collision. Recently Silber et al. (2010), using scale models of ships and whales in experimental flow tanks, demonstrated that under certain conditions reduced ship speed may reduce the probability of a collision given a severe encounter, and Vanderlaan and Taggart (2007) demonstrated that in the event of a collision, reduced ship speed may reduce the probability of severe injury or death. Our study is the first to demonstrate that the average distance between ships and whales decreases with increasing ship speed, confirming that speed reduction is an effective means for reducing the probability of severe encounters. If we assume that the closer that whales encounter ships the more likely they are to be struck, then, by extension, reducing ship speed may be an effective management action in reducing the probability of a collision.

It is beyond the scope of our study to identify the mechanism underlying the speed–distance relationship, although we feel the results are not likely a function of ship avoidance behavior of the whales. Cruise ships in our study area have no designated marine mammal observer on the bridge, which can influence the probability of detecting whales (Weinrich et al. 2010) needed prior to initiating avoidance measures. In addition, we regularly communicated with the ship captains/pilots and many have stated that when a whale is detected they generally retain course and speed, noting that the confined space within Glacier Bay hinders large course alterations. What is more, despite nearly 900 encounters with whales over four years we know of only two instances when ships altered course or speed in order to avoid whales (in both cases it was a group of >2 whales), and one instance when the ship captain knew of an approaching whale but took no evasive measures. In this instance, the observer radioed the ship bridge to inform them that a whale, which appeared to

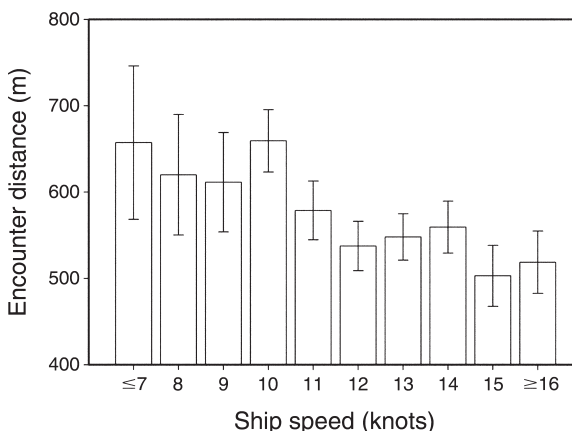


FIG. 5. Average encounter distances (mean $\pm$ SE) between cruise ships and humpback whales, binned for 1-knot speed increments in the Lower Glacier Bay subarea, 2006–2009. 1 knot = 0.5144 m/s.



PLATE 1. A humpback whale encounters a cruise ship in Glacier Bay National Park and Preserve. Photo credit: Janet Neilson, National Parks Service.

be resting at the surface, was directly in the path of the ship ~ 4 km away. Over the next nine minutes the observer repeatedly radioed the bridge as the ship–whale encounter distance decreased, and despite acknowledging the location of the whale, the ship captain/pilot maintained course and speed. As the ship approached to less than 70 m, the whale initiated a deep dive and a collision was narrowly avoided.

If in fact the speed–distance relationship is a function of whale avoidance behavior of the ships (as opposed to ship avoidance behavior of the whales), one scenario by which faster ships result in closer encounters could occur if whales are initiating avoidance behavior once an acoustical threshold is exceeded, and acoustic signals from ships are somewhat independent of speed. There is currently no information available regarding detection and response by large whales (cetaceans) to variation in received acoustic signals under different ship speeds (NRC 2005). Nevertheless, recent acoustic measurements of four cruise ships with diesel-electric propulsion systems recorded at the U.S. Navy's Southeast Alaska Acoustic Measurement Facility, including several ships used in our study, demonstrated that the overall sound levels emanating from these ships was similar at speeds differing by up to 8 knots (4.1 m/s) (Kipple 2002). If the acoustic signals are somewhat independent of speed, the distance between ship and whale at any acoustic threshold will thus be the same. Yet the whale will have

less time to process the signal and initiate avoidance measures for the faster ships. Using the data from Kipple (2002) as an example, if a whale initiated avoidance measures at 130dB (an arbitrarily defined threshold) it would have only 8 s before collision for a ship traveling 20 knots (10.3 m/s) vs. 16 s for ships traveling 10 knots (5.1 m/s).

Regardless of mechanism, our results add to the sparse yet growing evidence that reducing ship speed may serve as an effective measure in meeting whale conservation objectives. Ultimately however, application of our results will need confirmation in other areas with different oceanographic conditions, vessel types, and whale species. Nevertheless, we highlight the effectiveness of using shipboard observers for understanding the relationship between whales and ships, and testing the effectiveness of applied management actions.

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